BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

PLOT PLAN, ELEVATIONS AND PIPING ISOMETRICS

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DISCIPLINE : Piping

PROJECT : BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

STATUS : Issued for teaching (IFT)

Plot plan of the BFW steam heater unit with equipment coordinates and spacing rules, an elevation with the levels that set the P-101 suction head (static head, NPSH) and the gravity drain to ST-101, isometrics ISO-1 to ISO-5 of 6"-BW-101, 12"-LS-101 / 14"-LS-102, 3"-SC-101 / 4"-SC-102, 1"-VT-101 with the PSV-100 discharge and 3"-BW-102 with bills of materials, the walking route of the 3-D model and the layout checks. Drawn from layout.json.

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Numbers are taken from the BFW Steam Heater Ver 1 simulator model; values marked assumed / TBC / HOLD are not design data.

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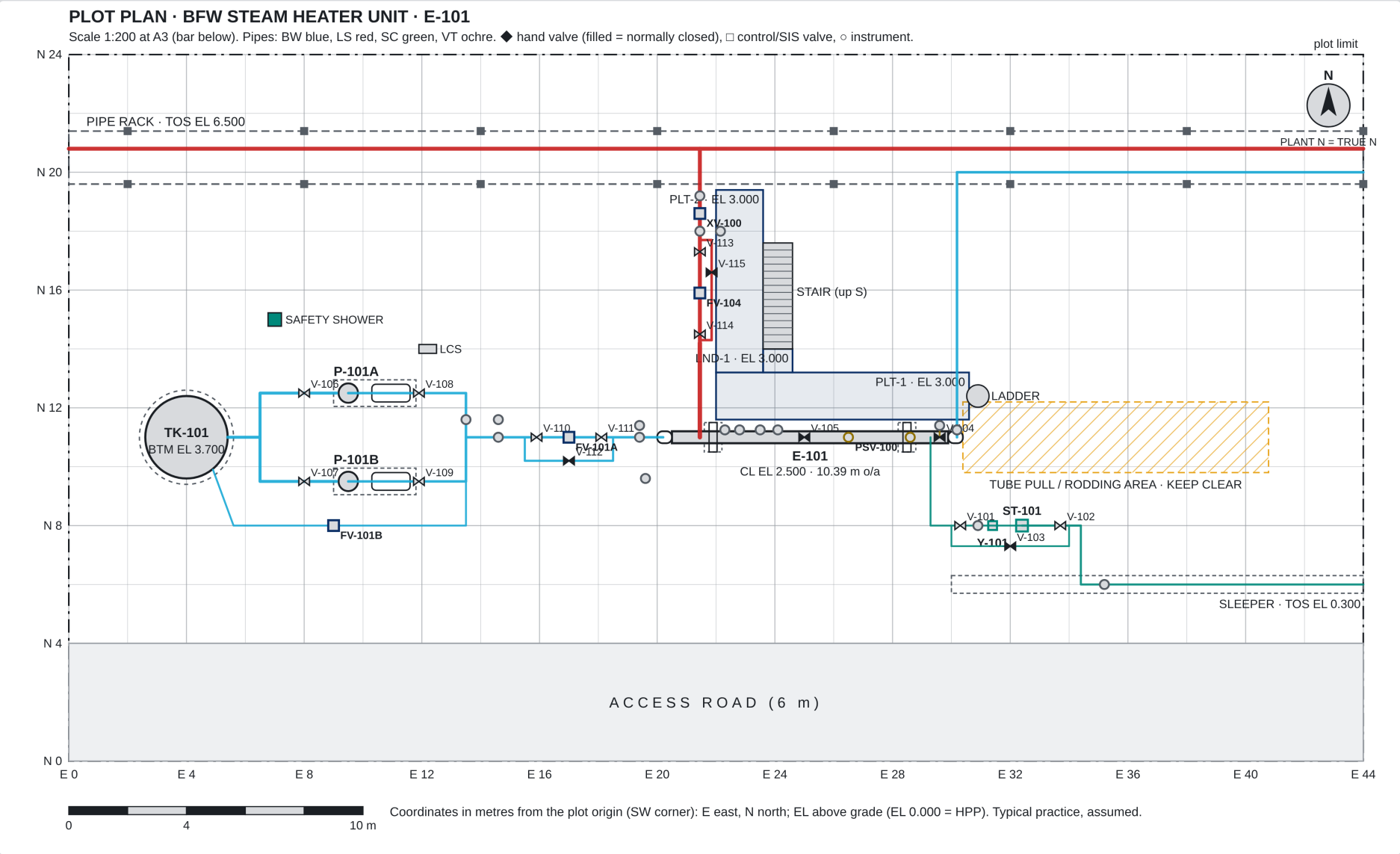
1 Purpose and basis

This document fixes where each item of the BFW steam heater unit stands and how each line is routed: the plot plan with coordinates, an elevation with the levels that set the pump suction head and the condensate drainage, and the piping isometrics of the main lines with their bills of materials. The P&ID BFWH-V1-PID-001 rev D gives the connections, line numbers and valves; BFWH-V1-ME-EQL-001 gives the equipment sizes; BFWH-V1-PE-HMB-001 gives the line sizes and the suction hydraulics.

All coordinates come from one file, layout.json (generated by p16/layout/mk_layout.py), which also builds the 3-D model of the Visual plant 3D tab. Units metres; E east, N north from the south-west corner of the plot; EL above grade (EL 0.000 = high point of paving). Positions, levels, spacing rules, support spans, piping material and schedules are **typical practice, assumed**; nothing here is for construction.

- Levels: grade EL 0.000; pump centreline EL 0.600; trap EL 0.500; E-101 centreline EL 2.500; platform deck EL 3.000; TK-101 bottom EL 3.700; rack top of steel EL 6.500.
- E-101 overall length 10.39 m = tube length 9.39 m (Design tab) + 2 × 0.05 m tubesheets + 2 × 0.45 m bonnets (BFWH-V1-ME-EQL-001 GA).
- Access (assumed): operated or read from a standing level up to 2.0 m above it and within 0.75 m of a deck edge; headroom over decks 2.1 m; condensate fall 1:100.
- Wind prevailing from the north-east (assumed); control room off-plot, about 60 m north (assumed).

2 Plot plan



Plot plan from layout.json, true scale (bar 0–10 m). BW blue, LS red, SC green, VT ochre; ◆ hand valve (filled = normally closed), □ control / SIS valve, ○ instrument. Positions typical practice, assumed.

3 Coordinates and spacing

Coordinates of the equipment and structures (centre and footprint). The Visual plant 3D tab reads the same values.

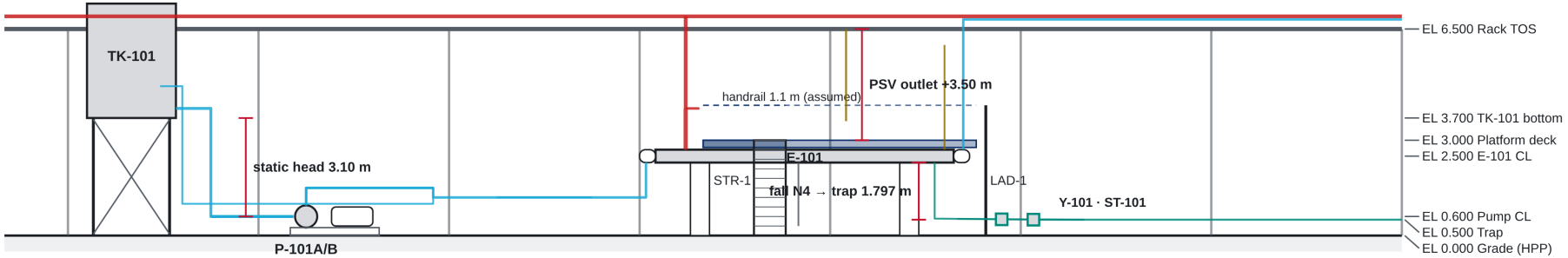
Tag	Item	E m	N m	Footprint E / N m	Levels	Size	Source
E-101	BFW steam heater (TEMA BEM, horizontal)	25.195	11.000	20.00–30.39 / 10.40–11.60	CL EL 2.500; shell bottom EL 2.297, top EL 2.703	10.39 m long (E 20.000 – 30.390), shell OD 406 mm	Design tab / BFWH-V1-ME-EQL-001
TK-101	BFW tank (20.6 m3)	4.000	11.000	2.40–5.60 / 9.40–12.60	bottom EL 3.700, top EL 7.300 (on stand)	Ø 2.8 × 3.6 m (assumed)	BFWH-V1-ME-EQL-001 (volume, NPSH); diameter/height assumed
P-101A	BFW pump A (motor 37 kW)	9.500	12.500	9.00–11.80 / 12.05–12.95	CL EL 0.600	2.80 × 0.90 m skid	BFWH-V1-EE-LL-001; skid size assumed
P-101B	BFW pump B, spare (motor 37 kW)	9.500	9.500	9.00–11.80 / 9.05–9.95	CL EL 0.600	2.80 × 0.90 m skid	BFWH-V1-EE-LL-001; skid size assumed
ST-101	Steam trap (float & thermostatic)	32.400	8.000	32.20–32.60 / 7.80–8.20	CL EL 0.500	in line	P&ID
Y-101	Strainer	31.400	8.000	31.25–31.55 / 7.85–8.15	CL EL 0.511	in line	P&ID
PLT-1	E-101 operating platform	26.300	12.400	22.00–30.60 / 11.60–13.20	deck EL 3.000	8.60 × 1.60 m	typical practice, assumed
PLT-2	Steam station walkway	22.800	16.300	22.00–23.60 / 13.20–19.40	deck EL 3.000	1.60 × 6.20 m	typical practice, assumed
LND-1	Stair-head landing	24.100	13.600	23.60–24.60 / 13.20–14.00	deck EL 3.000	1.00 × 0.80 m	typical practice, assumed
STR-1	Stair, grade to EL 3.000	24.100	15.800	23.60–24.60 / 14.00–17.60	EL 0.000 → 3.000	1.00 × 3.60 m	typical practice, assumed
LAD-1	Caged ladder (second escape)	30.900	12.400	30.50–31.30 / 12.00–12.80	EL 0.000 → 3.000	caged	typical practice, assumed
RACK	Pipe rack	22.000	20.500	0.00–44.00 / 19.60–21.40	TOS EL 6.500	44.00 × 1.80 m	typical practice, assumed
SLP-1	Condensate return sleeper	37.000	6.000	30.00–44.00 / 5.70–6.30	TOS EL 0.300	14.00 × 0.60 m	typical practice, assumed
ROAD	Access road (6 m)	22.000	2.000	0.00–44.00 / 0.00–4.00	EL 0.000	44.00 × 4.00 m	typical practice, assumed
PULL	Tube-bundle pull area (keep clear)	35.585	11.000	30.39–40.78 / 9.80–12.20	at grade	10.39 × 2.40 m	typical practice, assumed

Spacing and access rules (typical practice) evaluated from the coordinates

Rule	Required	Measured	Basis	OK
Pump to its suction vessel (TK-101): short, straight suction	≤ 6 m	plan 3.99 m (TK-101 outlet → P-101A suction); suction line 8.60 m	typical practice, assumed	✓
Clear access aisle along the pump row	≥ 1.0 m	2.10 m between the skids	typical practice, assumed	✓
Tube-bundle pull area behind E-101	≥ 10.39 m (tubes 9.39 + 1)	13.61 m from the rear bonnet to the plot limit; clear at tube level: yes	typical practice; fixed tubesheet: rodding and tube replacement	✓
Two means of escape from the platform	≥ 2	2 (STR-1, LAD-1), 6.3 m apart	typical practice	✓
Relief and vent outlets above the platform	≥ 3 m above deck	PSV-100 +3.5 m, VT-101 +3.0 m above the deck	typical practice, assumed	✓
Gravity drain to the trap, no lift after it	trap below N4; SC-102 not rising	N4 EL 2.297 − ST-101 EL 0.500 = 1.797 m; largest rise after the trap 0.000 m	steam-trap practice	✓
Access road to the pumps for maintenance	≤ 15 m	5.05 m from the road edge to the pump skids	typical practice, assumed	✓

4 Elevation and levels

ELEVATION LOOKING NORTH · levels and gravity heads



True scale both ways. All pipes projected onto the east–west plane; P-101B is behind P-101A. Levels from layout.json (E-101 CL, platform, rack and TK-101 bottom assumed; trap below N4 for gravity drainage).

Elevation looking north, true scale; all pipes projected on the east–west plane; P-101B is behind P-101A.

Static head. $z = EL_{TK\text{-}101\ bottom} - EL_{pump\ CL} = 3.700 - 0.600 =$ **3.100 m**; $\rho gz = 998 \times 9.81 \times 3.100 / 1000 =$ **30.35 kPa** ≥ K.Ptk 30 kPa (model suction basis; minimum level 3.063 m in BFWH-V1-ME-EQL-001) ✓.

NPSH available. 8"-BW-100 8.60 m (ID 202.7 mm), ΣK = 2 (entrance, valve, strainer) + 2 × 0.3 (elbows) + 1.0 (tee) = 3.60; f 0.0173; $h_s = (0.0173 \times 8.60/0.2027 + 3.60) \times 0.953^2/19.62$ = 0.200 m. NPSHa = 10.34 + 3.100 − 0.210 − 0.200 = **13.03 m** at 110.7 m³/h, 12.98 m at the rated 123.6 m³/h ≥ 1.3 × NPSHr 4.0 m (assumed, HOLD) = 5.2 m ✓. BFWH-V1-PE-HMB-001 (25 m equivalent length, z 3.063 m) gave 13.01 m: the process basis is conservative.

Gravity drain. Fall N4 → ST-101 = 2.297 − 0.500 = **1.797 m**, condensate head 942 × 9.81 × 1.797/1000 = **16.6 kPa** at the trap inlet (ρ at 121.3 °C). Trap ΔP at design 105 + 16.6 − 30 = 91.6 kPa. Run legs 3.00 m EL 0.565 → 0.535 (1.00 %); 3.11 m EL 0.535 → 0.500 (1.13 %) ≥ 1:100 ✓ (EQ-70 closed by Layout Rev 0b). No rise after the trap (0.000 m) ✓.

5 Access to valves and gauges

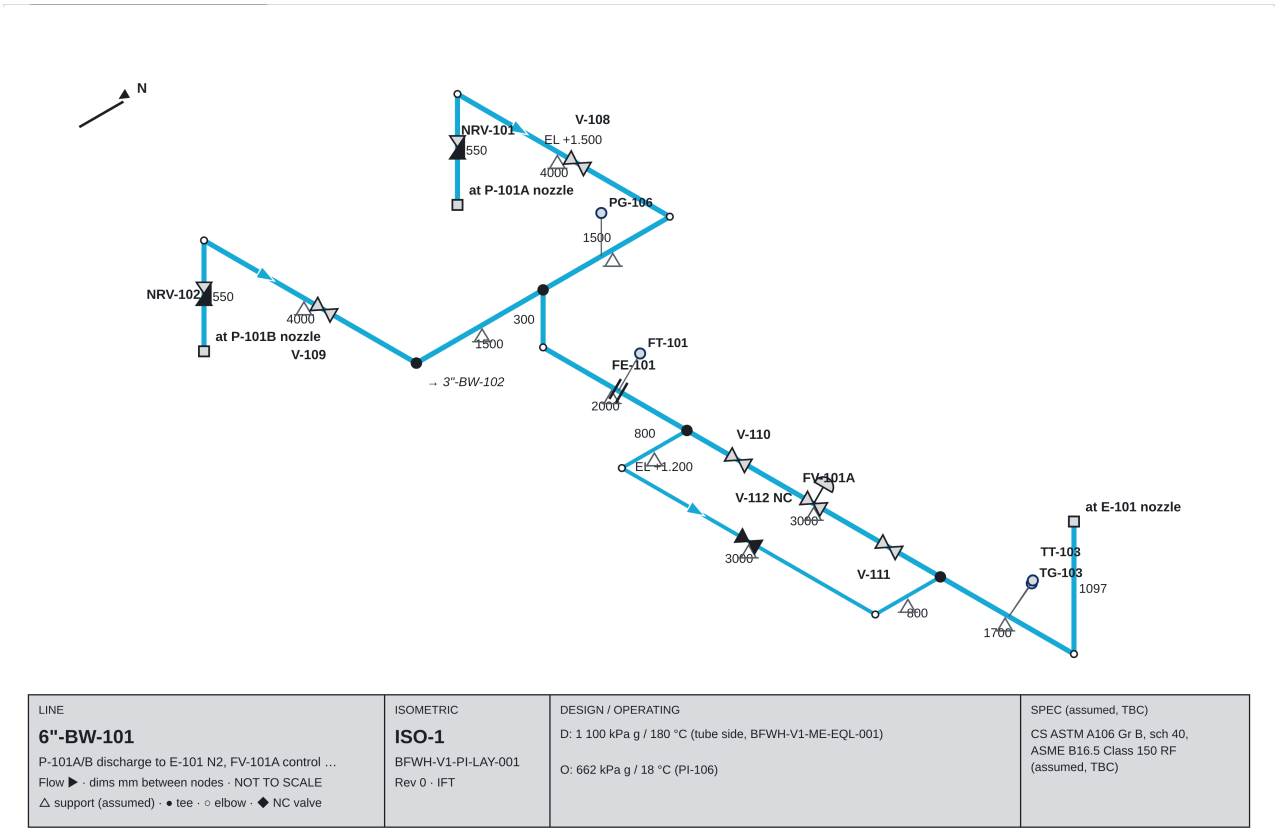
Each hand valve and gauge is operated or read from grade or a platform deck: at most 2.0 m above the standing level and within 0.75 m of the deck edge (typical practice, assumed).

Tag	What	EL	From	Height m	OK
V-101	hand valve, Trap inlet block	0.52	grade	0.52	✓
V-102	hand valve, Trap outlet block	0.50	grade	0.50	✓
V-103	hand valve, Trap bypass	0.51	grade	0.51	✓
V-104	hand valve, E-101 air vent	3.40	PLT-1 (EL 3.0)	0.40	✓
V-105	hand valve, E-101 shell drain	1.40	grade	1.40	✓
V-106	hand valve, P-101A suction	0.60	grade	0.60	✓
V-107	hand valve, P-101B suction	0.60	grade	0.60	✓
V-108	hand valve, P-101A discharge	1.50	grade	1.50	✓
V-109	hand valve, P-101B discharge	1.50	grade	1.50	✓
V-110	hand valve, FV-101A upstream block	1.20	grade	1.20	✓
V-111	hand valve, FV-101A downstream block	1.20	grade	1.20	✓
V-112	hand valve, FV-101A bypass (globe)	1.20	grade	1.20	✓
V-113	hand valve, FV-104 upstream block	4.00	PLT-2 (EL 3.0)	1.00	✓
V-114	hand valve, FV-104 downstream block	4.00	PLT-2 (EL 3.0)	1.00	✓
V-115	hand valve, FV-104 bypass (globe)	4.00	PLT-2 (EL 3.0)	1.00	✓
PG-106	gauge, P-101 discharge pressure (local; PI-106 on DCS)	1.80	grade	1.80	✓
FT-101	transmitter, BFW flow (FIC-101, FSLL-101 shared)	1.60	grade	1.60	✓
FV-101A	control valve, BFW flow control valve, FO	1.20	grade	1.20	✓
FV-101B	control valve, Minimum-flow recycle valve	1.00	grade	1.00	✓
TG-103	thermometer, BFW inlet thermometer (local)	1.20	grade	1.20	✓
TT-103	transmitter, BFW inlet temperature (TI-103)	1.40	grade	1.40	✓
PDT-109	transmitter, Tube-side differential pressure (PDI-109)	1.20	grade	1.20	✓
PI-105	transmitter, Flash steam header pressure (PT-105)	4.00	PLT-2 (EL 3.0)	1.00	✓
XV-100	SIS valve, SIS steam shut-off valve, FC (I-100)	4.00	PLT-2 (EL 3.0)	1.00	✓
FT-104	transmitter, Steam flow (FIC-104)	4.30	PLT-2 (EL 3.0)	1.30	✓
FV-104	control valve, Steam flow control valve, FC (stem position local)	4.00	PLT-2 (EL 3.0)	1.00	✓
PT-100	transmitter, Shell pressure (PI-100, PSHH-100 shared)	3.40	PLT-1 (EL 3.0)	0.40	✓
PG-100	gauge, Shell pressure gauge (local)	3.40	PLT-1 (EL 3.0)	0.40	✓
TI-100	transmitter, Steam temperature at the inlet (TI-100)	3.30	PLT-1 (EL 3.0)	0.30	✓
TG-100	thermometer, Steam thermometer (local)	3.30	PLT-1 (EL 3.0)	0.30	✓
PSV-100	relief valve, Shell relief valve, set TBC	3.30	PLT-1 (EL 3.0)	0.30	✓
VB-101	vacuum breaker, Vacuum breaker	3.00	PLT-1 (EL 3.0)	0.00	✓
TI-102	transmitter, Vent-line temperature (rev C, with AAV-101)	5.00	PLT-1 (EL 3.0)	2.00	✓
TT-101	transmitter, BFW outlet temperature (TIC-101)	4.00	PLT-1 (EL 3.0)	1.00	✓
TT-108	transmitter, BFW outlet temperature (TIAH/TSHH-108)	4.40	PLT-1 (EL 3.0)	1.40	✓
TG-101	thermometer, BFW outlet thermometer (local)	3.60	PLT-1 (EL 3.0)	0.60	✓
TG-107	thermometer, Condensate thermometer (local, upstream ST-101)	0.80	grade	0.80	✓
PG-110	gauge, Condensate return pressure (local)	0.90	grade	0.90	✓

No pipe runs over a platform deck with less than 2.1 m clear headroom (EQ-71 closed by Layout Rev 0b).

6 ISO-1 · 6"-BW-101

P-101A/B discharge to E-101 N2, FV-101A control station. Design: 1 100 kPa g / 180 °C (tube side, BFWH-V1-ME-EQL-001). Operating: 662 kPa g / 18 °C (PI-106). Pipe: 6" 20.20 m, 4" 4.60 m; 7 elbows, 4 tees, 0 reducer(s), 21 flanged joints, 10 supports.



ISO-1: isometric from the layout.json polylines, not to scale; dimensions in mm between nodes from the coordinates; supports and NRV positions assumed.

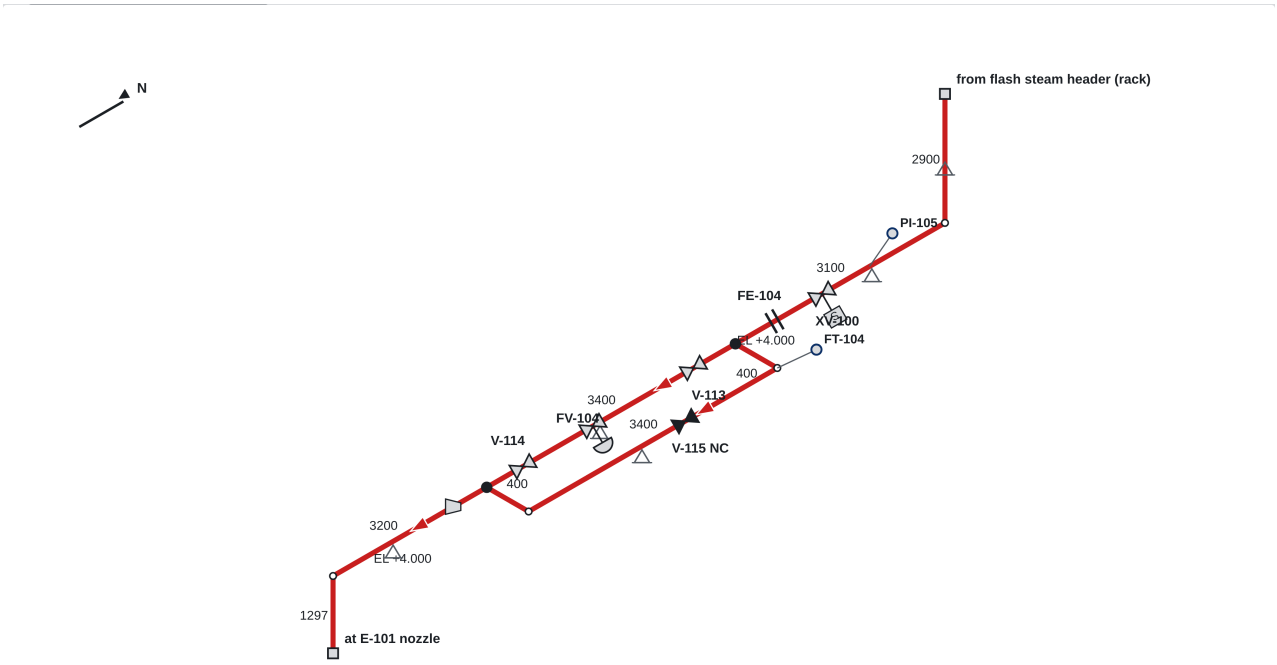
6.1 ISO-1 bill of materials

Item	Size	Qty	Note
Pipe	6"	20.20 m	CS A106 Gr B sch 40 (assumed, TBC); length from the layout polyline
Pipe	4"	4.60 m	CS A106 Gr B sch 40 (assumed, TBC); length from the layout polyline
Elbow, long radius 90°	6"	5	butt-weld (assumed)
Elbow, long radius 90°	4"	2	butt-weld (assumed)
Tee, equal	6"	1	butt-weld (assumed)
Tee, reducing	6" × 3"	1	butt-weld (assumed)
Tee, reducing	6" × 4"	2	butt-weld (assumed)
Branch connections to other lines	—	1	3"-BW-102 (counted in the tees above)
Gate valve V-108	6"	1	P-101A discharge · normally open · Class 150 flanged (assumed)
Gate valve V-109	6"	1	P-101B discharge · normally open · Class 150 flanged (assumed)
Gate valve V-110	6"	1	FV-101A upstream block · normally open · Class 150 flanged (assumed)
Gate valve V-111	6"	1	FV-101A downstream block · normally open · Class 150 flanged (assumed)
Globe valve V-112	4"	1	FV-101A bypass (globe) · normally closed · Class 150 flanged (assumed)
Control valve FV-101A	6"	1	BFW flow control valve, FO · Class 150 flanged (assumed)
Check valve NRV-101	6"	1	Check valve (P&ID; position assumed) · Class 150 flanged (assumed)
Check valve NRV-102	6"	1	Check valve (P&ID; position assumed) · Class 150 flanged (assumed)
Orifice flange pair + plate FE-101	6"	1	Class 300 orifice flanges (typical, assumed); straight runs TBC
Instrument connection (¾" root valve)	¾"	4	PG-106, FT-101, TG-103, TT-103 (assumed)
Flanged joints (pipe flange + gasket + bolt set)	line size	21	2 per valve / orifice / trap / strainer, 1 per equipment nozzle (E-101, P-101A, P-101B)
Pipe supports	—	10	spans 4" 4.3 m, 6" 5.2 m (typical, assumed); shoe / rest

Spec: CS ASTM A106 Gr B, sch 40, ASME B16.5 Class 150 RF (assumed, TBC).

7 ISO-2 · 12"-LS-101 / 14"-LS-102

Flash steam from the header to E-101 N1, XV-100 / FV-104 station. Design: header design TBC (maximum 300 kPa g assumed); LS-102: 350 kPa g / 180 °C (shell, BFWH-V1-ME-EQL-001). Operating: 200 kPa g header (PI-105 195 at the valve) → 105 kPa g / 121.3 °C (PI-100). Pipe: 14" 4.50 m, 12" 9.50 m, 6" 4.20 m; 4 elbows, 2 tees, 1 reducer(s), 13 flanged joints, 5 supports.



LINE	ISOMETRIC	DESIGN / OPERATING	SPEC (assumed, TBC)
12"-LS-101 / 14"-LS-102 Flash steam from the header to E-101 N1, XV-100 … Flow ▶ · dims mm between nodes · NOT TO SCALE △ support (assumed) · ● tee · ○ elbow · ◆ NC valve	**ISO-2** BFWH-V1-PI-LAY-001 Rev 0 · IFT	D: header design TBC (maximum 300 kPa g assumed); LS-102: 350 kPa g / 180 °C (shell, BFWH-V1-ME-EQL-001) O: 200 kPa g header (PI-105 195 at the valve) → 105 kPa g / 121.3 °C (PI-100)	CS ASTM A106 Gr B, sch 40, Class 150 RF, insulated (assumed, TBC)

ISO-2: isometric from the layout.json polylines, not to scale; dimensions in mm between nodes from the coordinates; supports and NRV positions assumed.

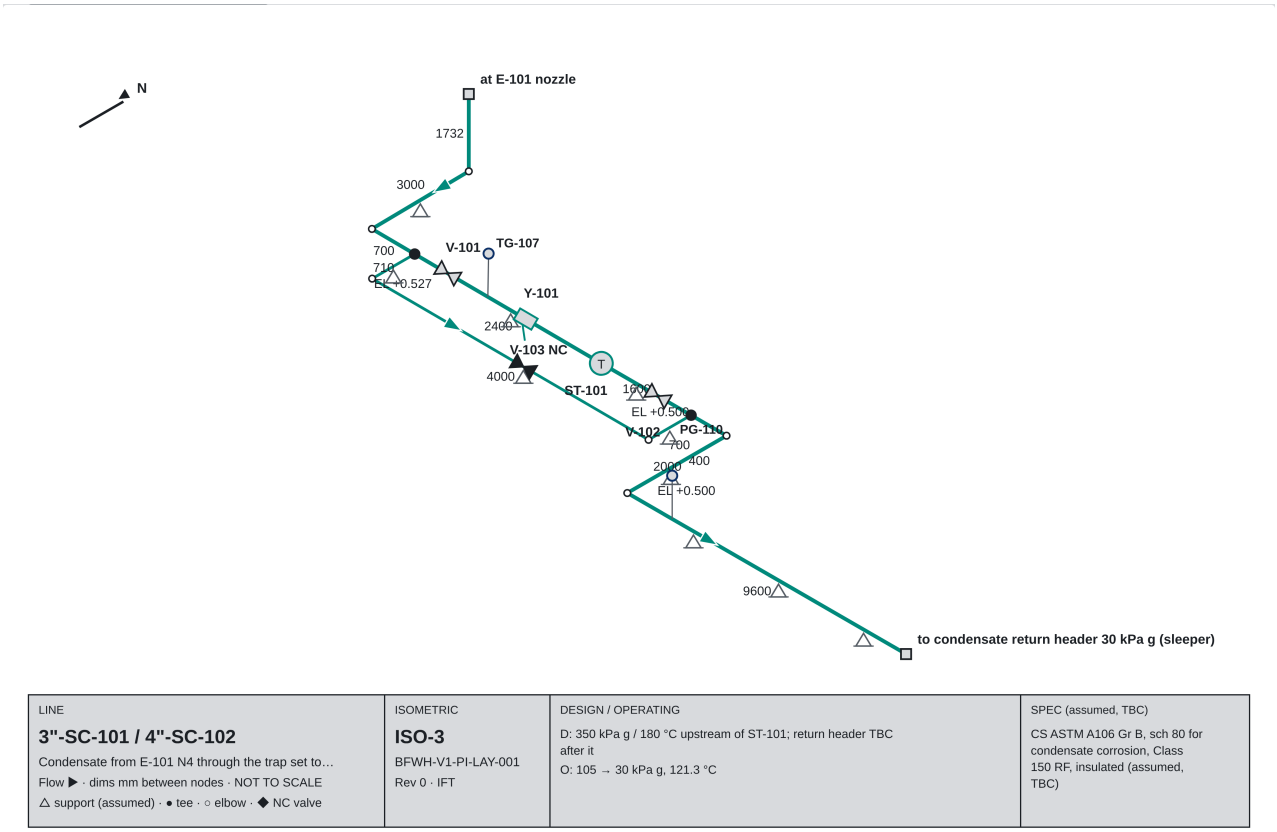
7.1 ISO-2 bill of materials

Item	Size	Qty	Note
Pipe	14"	4.50 m	CS A106 Gr B sch 40 (assumed, TBC); length from the layout polyline
Pipe	12"	9.50 m	CS A106 Gr B sch 40 (assumed, TBC); length from the layout polyline
Pipe	6"	4.20 m	CS A106 Gr B sch 40 (assumed, TBC); length from the layout polyline
Elbow, long radius 90°	14"	1	butt-weld (assumed)
Elbow, long radius 90°	12"	1	butt-weld (assumed)
Elbow, long radius 90°	6"	2	butt-weld (assumed)
Tee, reducing	12" × 6"	2	butt-weld (assumed)
Reducer, concentric	14" × 12"	1	at E 21.45 N 14.20
Gate valve V-113	12"	1	FV-104 upstream block · normally open · Class 150 flanged (assumed)
Gate valve V-114	12"	1	FV-104 downstream block · normally open · Class 150 flanged (assumed)
Globe valve V-115	6"	1	FV-104 bypass (globe) · normally closed · Class 150 flanged (assumed)
Shut-off valve (SIS) XV-100	12"	1	SIS steam shut-off valve, FC (I-100) · Class 150 flanged (assumed)
Control valve FV-104	12"	1	Steam flow control valve, FC (stem position local) · Class 150 flanged (assumed)
Orifice flange pair + plate FE-104	12"	1	Class 300 orifice flanges (typical, assumed); straight runs TBC
Instrument connection (¾" root valve)	¾"	2	PI-105, FT-104 (assumed)
Flanged joints (pipe flange + gasket + bolt set)	line size	13	2 per valve / orifice / trap / strainer, 1 per equipment nozzle (E-101)
Pipe supports	—	5	spans 6" 5.2 m, 12" 7.0 m, 14" 7.6 m (typical, assumed); guide, shoe / rest

Spec: CS ASTM A106 Gr B, sch 40, Class 150 RF, insulated (assumed, TBC).

8 ISO-3 · 3"-SC-101 / 4"-SC-102

Condensate from E-101 N4 through the trap set to the return header. Design: 350 kPa g / 180 °C upstream of ST-101; return header TBC after it. Operating: 105 → 30 kPa g, 121.3 °C. Pipe: 4" 13.60 m, 3" 7.84 m, 2" 5.40 m; 6 elbows, 2 tees, 1 reducer(s), 11 flanged joints, 10 supports.



LINE	ISOMETRIC	DESIGN / OPERATING	SPEC (assumed, TBC)
3"-SC-101 / 4"-SC-102 Condensate from E-101 N4 through the trap set to… Flow ▶ · dims mm between nodes · NOT TO SCALE △ support (assumed) · ● tee · ○ elbow · ◆ NC valve	**ISO-3** BFWH-V1-PI-LAY-001 Rev 0 · IFT	D: 350 kPa g / 180 °C upstream of ST-101; return header TBC after it O: 105 → 30 kPa g, 121.3 °C	CS ASTM A106 Gr B, sch 80 for condensate corrosion, Class 150 RF, insulated (assumed, TBC)

ISO-3: isometric from the layout.json polylines, not to scale; dimensions in mm between nodes from the coordinates; supports and NRV positions assumed.

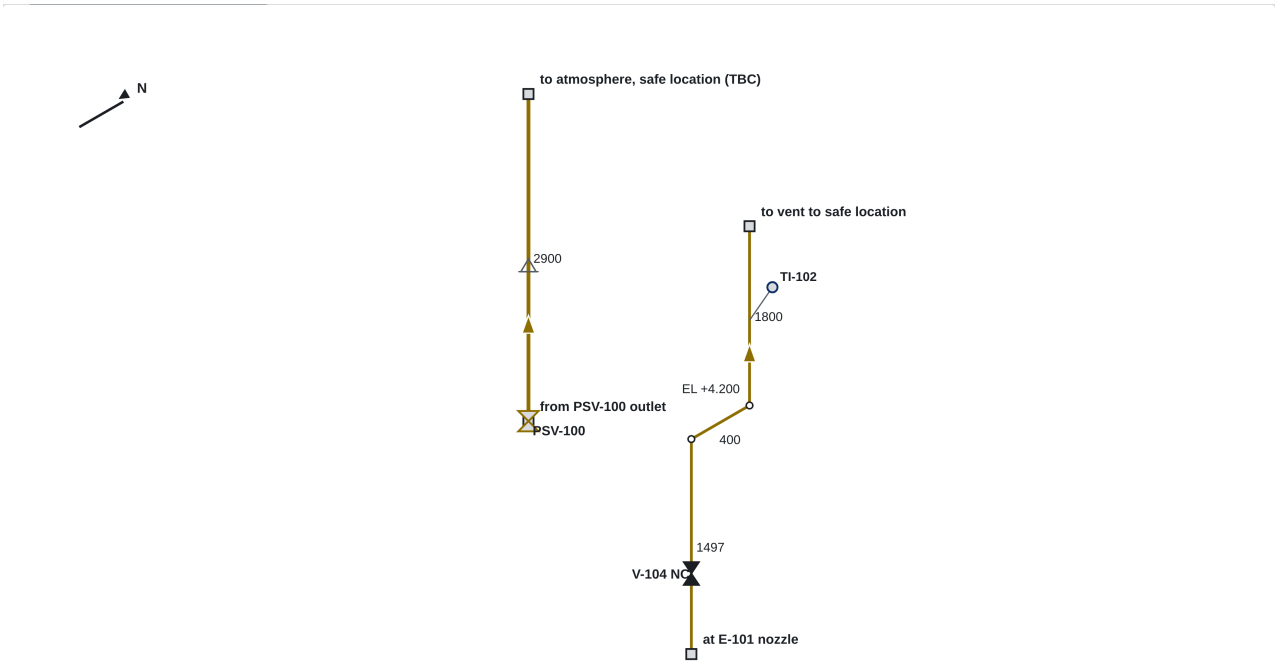
8.1 ISO-3 bill of materials

Item	Size	Qty	Note
Pipe	4"	13.60 m	CS A106 Gr B sch 80 (assumed, TBC); length from the layout polyline
Pipe	3"	7.84 m	CS A106 Gr B sch 80 (assumed, TBC); length from the layout polyline
Pipe	2"	5.40 m	CS A106 Gr B sch 80 (assumed, TBC); length from the layout polyline
Elbow, long radius 90°	4"	2	butt-weld (assumed)
Elbow, long radius 90°	3"	2	butt-weld (assumed)
Elbow, long radius 90°	2"	2	butt-weld (assumed)
Tee, reducing	3" × 2"	1	butt-weld (assumed)
Tee, reducing	4" × 2"	1	butt-weld (assumed)
Reducer, concentric	4" × 3"	1	at E 32.40 N 8.00
Gate valve V-101	3"	1	Trap inlet block · normally open · Class 150 flanged (assumed)
Gate valve V-102	4"	1	Trap outlet block · normally open · Class 150 flanged (assumed)
Globe valve V-103	2"	1	Trap bypass · normally closed · Class 150 flanged (assumed)
Steam trap ST-101	3"	1	flanged (assumed); datasheet BFWH-V1-ME-EQL-001
Y-strainer Y-101	3"	1	flanged (assumed); datasheet BFWH-V1-ME-EQL-001
Instrument connection (¾" root valve)	¾"	2	TG-107, PG-110 (assumed)
Flanged joints (pipe flange + gasket + bolt set)	line size	11	2 per valve / orifice / trap / strainer, 1 per equipment nozzle (E-101)
Pipe supports	—	10	spans 2" 3.0 m, 3" 3.7 m, 4" 4.3 m (typical, assumed); guide, rest on sleeper / grade support

Spec: CS ASTM A106 Gr B, sch 80 for condensate corrosion, Class 150 RF, insulated (assumed, TBC).

9 ISO-4 · 1"-VT-101 + PSV-100 discharge

E-101 air vent and relief valve discharge to a safe location. Design: 350 kPa g / 180 °C (shell); PSV outlet: atmospheric, HOLD. Operating: 105 kPa g at V-104 (normally closed). Pipe: 4" 2.90 m, 1" 3.70 m; 2 elbows, 0 tees, 0 reducer(s), 4 flanged joints, 1 supports.



LINE	ISOMETRIC	DESIGN / OPERATING	SPEC (assumed, TBC)
1"-VT-101 + PSV-100 discharge E-101 air vent and relief valve discharge to a s… Flow ▶ · dims mm between nodes · NOT TO SCALE △ support (assumed) · ● tee · ○ elbow · ◆ NC valve	**ISO-4** BFWH-V1-PI-LAY-001 Rev 0 · IFT	D: 350 kPa g / 180 °C (shell); PSV outlet: atmospheric, HOLD O: 105 kPa g at V-104 (normally closed)	CS ASTM A106 Gr B, sch 80 small bore, Class 150 RF (assumed, TBC)

ISO-4: isometric from the layout.json polylines, not to scale; dimensions in mm between nodes from the coordinates; supports and NRV positions assumed.

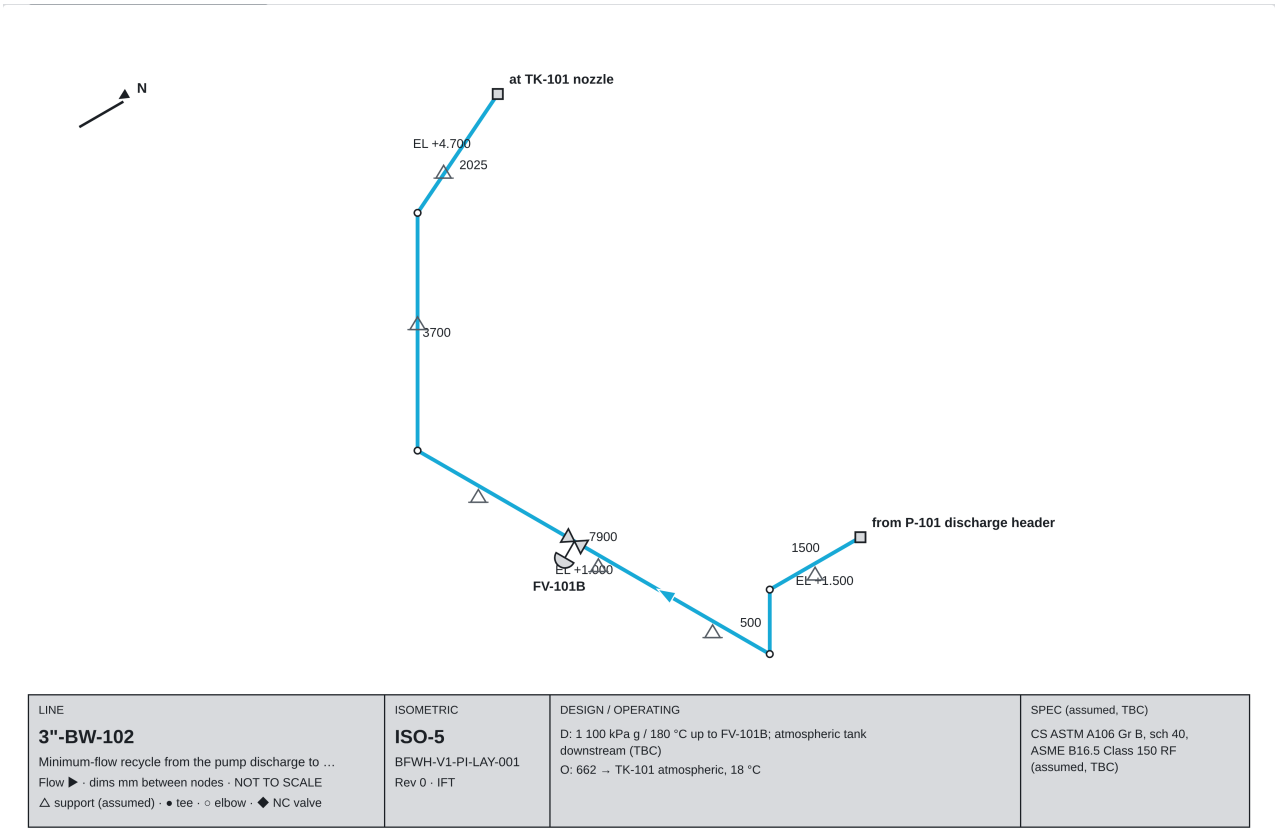
9.1 ISO-4 bill of materials

Item	Size	Qty	Note
Pipe	4"	2.90 m	CS A106 Gr B sch 80 (assumed, TBC); length from the layout polyline
Pipe	1"	3.70 m	CS A106 Gr B sch 80 (assumed, TBC); length from the layout polyline
Elbow, long radius 90°	1"	2	butt-weld (assumed)
Globe valve V-104	1"	1	E-101 air vent · normally closed · Class 150 flanged (assumed)
Relief valve PSV-100 (outlet connection)	HOLD	1	size and set from BFWH-V1-PE-RLB-001 — HOLD
Instrument connection (¾" root valve)	¾"	1	TI-102 (assumed)
Flanged joints (pipe flange + gasket + bolt set)	line size	4	2 per valve / orifice / trap / strainer, 1 per equipment nozzle (E-101, PSV-100)
Pipe supports	—	1	spans 1" 2.1 m, 4" 4.3 m (typical, assumed); guide

Spec: CS ASTM A106 Gr B, sch 80 small bore, Class 150 RF (assumed, TBC).

10 ISO-5 · 3"-BW-102

Minimum-flow recycle from the pump discharge to TK-101 through FV-101B. Design: 1 100 kPa g / 180 °C up to FV-101B; atmospheric tank downstream (TBC). Operating: 662 → TK-101 atmospheric, 18 °C. Pipe: 3" 15.62 m; 4 elbows, 0 tees, 0 reducer(s), 3 flanged joints, 6 supports.



ISO-5: isometric from the layout.json polylines, not to scale; dimensions in mm between nodes from the coordinates; supports and NRV positions assumed.

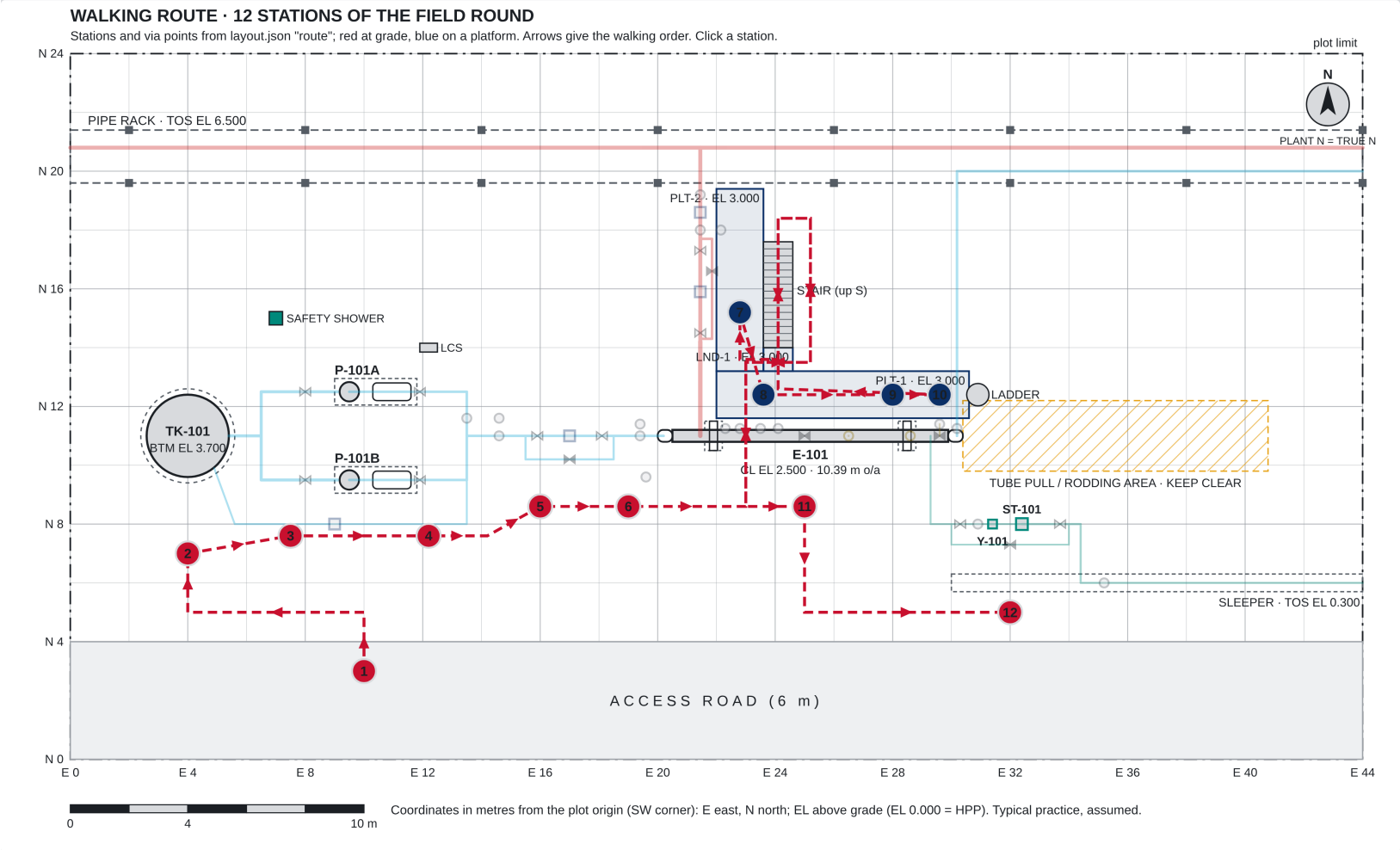
10.1 ISO-5 bill of materials

Item	Size	Qty	Note
Pipe	3"	15.62 m	CS A106 Gr B sch 40 (assumed, TBC); length from the layout polyline
Elbow, long radius 90°	3"	4	butt-weld (assumed)
Control valve FV-101B	3"	1	Minimum-flow recycle valve · Class 150 flanged (assumed)
Flanged joints (pipe flange + gasket + bolt set)	line size	3	2 per valve / orifice / trap / strainer, 1 per equipment nozzle (TK-101)
Pipe supports	—	6	spans 3" 3.7 m (typical, assumed); guide, shoe / rest

Spec: CS ASTM A106 Gr B, sch 40, ASME B16.5 Class 150 RF (assumed, TBC).

11 From layout to the 3-D model

layout.json is the single source of these drawings and of the 3-D scene of the Visual plant 3D tab. The outside operator walks 12 stations; red at grade, blue on a platform; the path follows the walkway (via) points of the 3-D walk: under E-101 at E 23.0 at grade, up STR-1 to the landing LND-1 before station 7 and down again before station 11.



Walking route of the field round on the plot plan.

11.1 layout.json and the stations

Key	Count	Holds
equipment	6	tag, kind, position, footprint, levels, source
structures	13	platforms, stair, ladder, rack, pedestals, sleeper, road, pull zone, LCS, shower
pipes	15	line number, service, size, from / to, polyline (and branch)
hand_valves	15	tag, model key, normal NO / NC, position
instruments	25	tag, kind, position, DCS tag
route	12	stations with position, standing level and items
spacing_rules	7	rule, requirement, measurement, basis

Route stations

No.	Station	E, N m	On	Items
1	Unit entrance (south road)	10.0, 3.0	grade	—
2	TK-101 BFW tank	4.0, 7.0	grade	TK-101
3	Pump suction (V-106, V-107)	7.5, 7.6	grade	V-106, V-107, P-101A, P-101B
4	P-101A/B discharge	12.2, 7.6	grade	V-108, V-109, PG-106, FV-101B
5	FV-101A control station	16.0, 8.6	grade	FT-101, V-110, FV-101A, V-111, V-112
6	E-101 front end (BFW in)	19.0, 8.6	grade	TG-103, TT-103, PDT-109
7	Steam station (platform EL 3.0)	22.8, 15.2	PLT-2	PI-105, XV-100, FT-104, V-113, FV-104, V-114, V-115
8	Shell top: pressure and temperature	23.6, 12.4	PLT-1	PT-100, PG-100, TI-100, TG-100
9	Relief and vent end	28.0, 12.4	PLT-1	PSV-100, VB-101, V-104, TI-102
10	E-101 rear end (BFW out)	29.6, 12.4	PLT-1	TT-101, TT-108, TG-101
11	Shell drain (grade)	25.0, 8.6	grade	V-105
12	Trap station	32.0, 5.0	grade	V-101, TG-107, ST-101, V-102, V-103, PG-110

12 Checks

34 of 35 agree; the 1 ✗ is an open engineering query (EQ-72). EQ-70, EQ-71, EQ-73, EQ-74 were closed by layout revisions 0b / 0c and their checks pass. Computed in the simulator page from layout.json, the P&ID text, the model and the Design tab.

Check	This document	Model / other document	Result
Every equipment item of BFWH-V1-ME-EQL-001 and the PFD is on the plot plan	8 of 8: E-101, P-101A, P-101B, TK-101, Y-101, ST-101, VB-101, PSV-100	all placed (equipment or relief / vacuum device)	✓
ISO-1: line number(s) exist on BFWH-V1-PID-001 rev D	6"-BW-101	found in the P&ID text	✓
ISO-2: line number(s) exist on BFWH-V1-PID-001 rev D	12"-LS-101, 14"-LS-102	found in the P&ID text	✓
ISO-3: line number(s) exist on BFWH-V1-PID-001 rev D	3"-SC-101, 4"-SC-102	found in the P&ID text	✓
ISO-4: line number(s) exist on BFWH-V1-PID-001 rev D	1"-VT-101	found in the P&ID text	✓
ISO-5: line number(s) exist on BFWH-V1-PID-001 rev D	3"-BW-102	found in the P&ID text	✓
Every pipe drawn on an isometric carries a line number	10 of 11	no number: PSV-100 discharge	✗ EQ-72 raised
Pipe size in layout.json = size in the line number	9 numbered runs	all agree	✓
Every hand valve V-101…V-115 of the P&ID has a position on a pipe	15 of 15 placed; 15 on a pipe centreline (± 50 mm)	all on their lines	✓
Every hand valve tag is on BFWH-V1-PID-001 rev D	15 of 15	all found	✓
Hand-valve normal position (NO / NC) = the model at design	15 of 15 compared	all agree	✓
Every instrument tag in layout.json exists on the P&ID or in the model	25 of 25 (23 P&ID, 2 model local gauges)	all found	✓
Every in-line instrument (control valve, SIS valve, orifice) sits on its line	6 items	all on their lines	✓
FV-101A station: bypass tees outside the block valves V-110 / V-111 (P&ID topology)	along the line: tee 8.35 · V-110 8.75 · FV-101A 9.85 · V-111 10.95 · tee 11.35 m	BFWH-V1-PID-001 rev D; model: the bypass works with a block valve closed	✓

Check	This document	Model / other document	Result
FV-104 station: bypass tees outside the block valves V-113 / V-114 (P&ID topology)	along the line: tee 6.00 · V-113 6.40 · FV-104 7.80 · V-114 9.20 · tee 9.40 m	BFWH-V1-PID-001 rev D; model: the bypass works with a block valve closed	✓
ST-101 station: bypass tees outside the block valves V-101 / V-102 (P&ID topology)	along the line: tee 5.44 · V-101 5.74 · ST-101 7.84 · V-102 9.14 · tee 9.44 m	BFWH-V1-PID-001 rev D; model: the bypass works with a block valve closed	✓
Spacing: Pump to its suction vessel (TK-101): short, straight suction	plan 3.99 m (TK-101 outlet → P-101A suction); suction line 8.60 m	≤ 6 m (typical practice, assumed)	✓
Spacing: Clear access aisle along the pump row	2.10 m between the skids	≥ 1.0 m (typical practice, assumed)	✓
Spacing: Tube-bundle pull area behind E-101	13.61 m from the rear bonnet to the plot limit; clear at tube level: yes	≥ 10.39 m (tubes 9.39 + 1) (typical practice; fixed tubesheet: rodding and tube replacement)	✓
Spacing: Two means of escape from the platform	2 (STR-1, LAD-1), 6.3 m apart	≥ 2 (typical practice)	✓
Spacing: Relief and vent outlets above the platform	PSV-100 +3.5 m, VT-101 +3.0 m above the deck	≥ 3 m above deck (typical practice, assumed)	✓
Spacing: Gravity drain to the trap, no lift after it	N4 EL 2.297 − ST-101 EL 0.500 = 1.797 m; largest rise after the trap 0.000 m	trap below N4; SC-102 not rising (steam-trap practice)	✓
Spacing: Access road to the pumps for maintenance	5.05 m from the road edge to the pump skids	≤ 15 m (typical practice, assumed)	✓
E-101 length on the plan = Design tab tube length + 2 tubesheets + 2 bonnets (EQL GA)	10.390 m (E 20.000 – 30.390)	9.386 + 2 × 0.05 + 2 × 0.45 = 10.386 m (ENG.ds(), BFWH-V1-ME-EQL-001)	✓
E-101 N1 and N4 positions = EQL nozzle schedule	N1 0.95 m from the front tubesheet; N4 0.60 m from the rear tubesheet	EQL: N1 1.0 m (± 0.1), N4 0.6 m	✓
TK-101 static head ≥ model suction basis	z 3.100 m, ρgz 30.4 kPa	K.Ptk 30 kPa g = 3.063 m (EQL)	✓
NPSHa (layout suction, rated flow) ≥ 1.3 × NPSHr	12.98 m	1.3 × 4.0 m (assumed) = 5.20 m	✓
N4 above the trap (gravity) and no lift after the trap	fall 1.797 m; SC-102 rise 0.000 m	steam-trap practice	✓
3"-SC-101 run legs fall ≥ 1:100 toward the trap	3.00 m: EL 0.565 → 0.535 (1.00 %); 3.11 m: EL 0.535 → 0.500 (1.13 %)	1:100 over 6.11 m = 61 mm (typical practice)	✓ (EQ-70 closed)
Every hand valve and gauge can be reached from grade or a platform	38 of 38	within 2 m height / 0.75 m reach (typical practice)	✓ (EQ-73 closed)
Every bypass and branch end lands on its parent line (± 5 mm)	3 bypasses, 2 branches	all connected	✓ (EQ-74 closed)
Walkway headroom ≥ 2.1 m along the walking route at grade (stations and via points)	lowest 2.30 m under E-101 shell at E 23.0, N 11.0	typical practice, assumed (deck structure 0.2 m, pipe within OD/2 + 0.3 m of the path)	✓
Every station of the walking route stands on grade or a platform deck	12 of 12	all supported	✓
Clear headroom ≥ 2.1 m over platform decks	none below	typical practice, assumed	✓ (EQ-71 closed)
All equipment inside the plot limits	6 of 6	plot 0, 0, 44, 24	✓

13 Engineering queries

EQ	Subject	Query	Status
EQ-70	Fall of the 3"-SC-101 condensate line to ST-101	The layout routes 3"-SC-101 straight down from N4 (EL 2.297) to EL 0.500 and then 6.11 m level at EL 0.500 to the trap. The vertical drop gives the gravity head (1.80 m, about 16.6 kPa of condensate), but the level legs have no fall. Typical practice is a fall of at least 1:100 toward the trap (61 mm over 6.11 m) so condensate cannot lie in the line at low load or after a trip. Confirm the fall on the isometric (start the level legs at EL 0.561 at the riser) or accept a level line flooded up to the trap. Resolution: Layout Rev 0b lays 3"-SC-101 with a 1:100 fall (EL 0.565 → 0.535 → 0.500 at ST-101); the page check now passes.	Closed
EQ-71	Headroom over the E-101 platforms	The layout runs pipes over the operating platforms below the 2.1 m clear headroom of typical practice: 14"-LS-102 crosses PLT-1 at EL 3.600 (0.42 m clear above the deck, 1.62 m long), 6"-BW-103 crosses PLT-1 at EL 4.000 (0.92 m clear) and 12"-LS-101 with its FV-104 bypass runs along the PLT-2 walkway at EL 4.600 (1.44 m clear). LS-102 and BW-103 cross PLT-1 between the stair STR-1 and the ladder LAD-1, so a person between them must step over or duck under a pipe to reach either escape. Re-route at the next layout revision: cross the deck at 2.1 m clear or pass under it, and place the steam station beside the walkway instead of over it; then update the 3-D model. Resolution: Layout Rev 0b lowers the steam station to EL 4.0 beside a walkway east of the line (PLT-2 E 22.0–23.6), starts PLT-1 at E 22.0 so 14"-LS-102 drops west of it, and takes 6"-BW-103 straight to the rack (3.8 m clear over PLT-1); the page check finds no pipe below 2.1 m.	Closed
EQ-72	PSV-100 discharge line has no line number	The PSV-100 discharge (drawn 4", EL 3.6 to EL 6.5, open to atmosphere at a safe location) has no line number on BFWH-V1-PID-001 rev D, and neither has the 1" shell drain at V-105. Assign line numbers (for example 4"-VT-102 and 1"-DR-101), size the PSV discharge from the relief load (BFWH-V1-PE-RLB-001, PSV-100 size HOLD), and add a weep hole and the discharge reaction support to the isometric.	Open

EQ	Subject	Query	Status
EQ-73	V-115 (FV-104 bypass) out of reach	Layout Rev 0b puts the FV-104 bypass on the west side of 12"-LS-101 at E 20.8, EL 4.0. V-115 is then 1.2 m in plan from the PLT-2 deck edge (E 22.0) and 4.0 m above grade: it can be operated neither from the walkway (reach 0.75 m, typical practice) nor from grade. V-113 and V-114 on the main line (E 21.45, 0.55 m from the deck edge) are reachable. Bring the bypass within reach: for example run it in the vertical plane above the main line (EL 4.6, 1.6 m above the deck, 0.55 m from the edge), or add a chain wheel (operability of a throttling globe on a chain is poor). The bypass is used to keep steam on E-101 while FV-104 is maintained, so it must be operable. Resolution: Layout Rev 0c moves the FV-104 bypass to the walkway side (E 21.85, V-115 at N 16.6), within reach of PLT-2; the page access check now passes.	Closed
EQ-74	Trap bypass tee off the sloped 3"-SC-101	With the 1:100 fall of Layout Rev 0b, 3"-SC-101 is at EL 0.527 at E 30.0, but the V-103 trap bypass still starts at EL 0.500 there: the upstream tee is 27 mm below the line, so the bypass does not connect in the model and ISO-3 shows it as a loose end. Set the bypass start to the line elevation (EL 0.527) or start the bypass upstream of the first fall; check V-101 (EL 0.520 vs line 0.524) at the same time. Resolution: Layout Rev 0c starts the trap bypass at the SC-101 elevation (EL 0.527; V-101 0.524, Y-101 0.511, V-103 0.514); the connection check now passes and ISO-3 closes.	Closed

Notes

- Drawings are the simulator tab's own SVGs rendered at high resolution; all numbers come from layout.json and the Mechanical / Process tab data (data.json of the Layout & isometrics tab).
- Positions, levels, spacing rules, support spans, piping material and schedules are typical practice, assumed; piping class, PSV-100 size and NPSHr are HOLD.
- Do not use for construction.